

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1518

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I G. Jaswanth - 192472235 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Jaswanth

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning.
 Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
 Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
 Some applications must remain on-premises due to compliance requirements.
 Design a hybrid architecture that ensures secure communication between environments.
 Explain how identity management and encryption should be implemented.
 Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
 The system must continue operating even if one region goes down.
 Propose a multi-region deployment strategy in the cloud.
 Explain how data replication and failover mechanisms should be configured.
 Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Question 1

1. Scenario Understanding

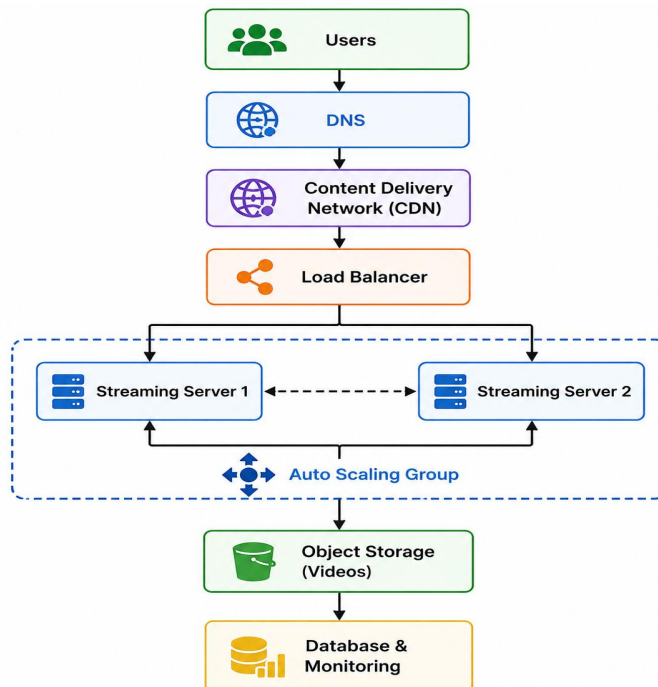
The media company plans to migrate its video streaming platform to the cloud. Since millions of users may watch videos simultaneously, the system must deliver content with **high availability, low latency, and minimal buffering**. The cloud architecture should automatically scale resources according to demand while ensuring efficient content delivery and cost optimization.

2. Challenges / Risks

The major challenges include:

- Sudden spikes in user traffic during popular events.
- Increased server load causing buffering and slow response.
- High bandwidth consumption.
- Single server failures affecting service availability.
- High operational costs if resources are over-provisioned.

3. Proposed Cloud Architecture



4. Scalability and Caching Techniques

Scalability

The platform should use **horizontal scaling**, where additional streaming servers are automatically added during peak demand and removed during low traffic.

Benefits:

- Supports millions of concurrent users.
- Prevents server overload.
- Improves availability.

Caching

A **Content Delivery Network (CDN)** caches frequently accessed videos at edge servers located close to users.

Benefits:

- Reduces latency.
- Minimizes buffering.
- Decreases bandwidth usage.
- Improves user experience.

5. Load Balancing and Auto-Scaling

Load Balancing

A load balancer distributes incoming user requests evenly among multiple streaming servers.

Advantages:

- Prevents individual server overload.
- Improves response time.
- Ensures high availability.
- Supports fault tolerance.

Auto Scaling

Auto Scaling monitors CPU usage, memory utilization, or network traffic and automatically adjusts the number of server instances.

For example:

- High demand → Launch additional servers.
- Low demand → Terminate unnecessary servers.

This maintains consistent performance while reducing costs.

6. Cost Optimization Strategies

To reduce cloud costs, the company should:

- Use **CDN caching** to minimize repeated data transfers.
- Enable **Auto Scaling** to allocate resources only when needed.

- Store videos in **low-cost object storage**.
- Move rarely accessed videos to archival storage.
- Monitor resource utilization and remove idle instances.
- Use reserved or savings plans for predictable workloads and on-demand instances for traffic spikes.

7. Benefits

The proposed architecture provides:

- High availability.
- Scalability for millions of users.
- Reduced buffering through CDN caching.
- Fault tolerance using load balancing.
- Automatic resource management with Auto Scaling.
- Lower operational costs through optimized cloud resource usage.

8. Justification

The proposed cloud architecture combines **CDN caching, load balancing, horizontal auto-scaling, object storage, and monitoring** to deliver a reliable and scalable video streaming platform. These technologies ensure smooth streaming during peak demand while optimizing resource utilization and minimizing operational costs. Therefore, the solution effectively meets the company's requirements for performance, availability, scalability, and cost efficiency.

Question 2

1. Scenario Understanding

The fintech startup is deploying its containerized applications across multiple cloud providers to improve availability, avoid vendor lock-in, and ensure business continuity. However, managing containers consistently across different cloud platforms presents challenges such as deployment consistency, networking, monitoring, and service communication. A centralized orchestration platform is required to automate deployment, scaling, and management.

2. Challenges / Risks

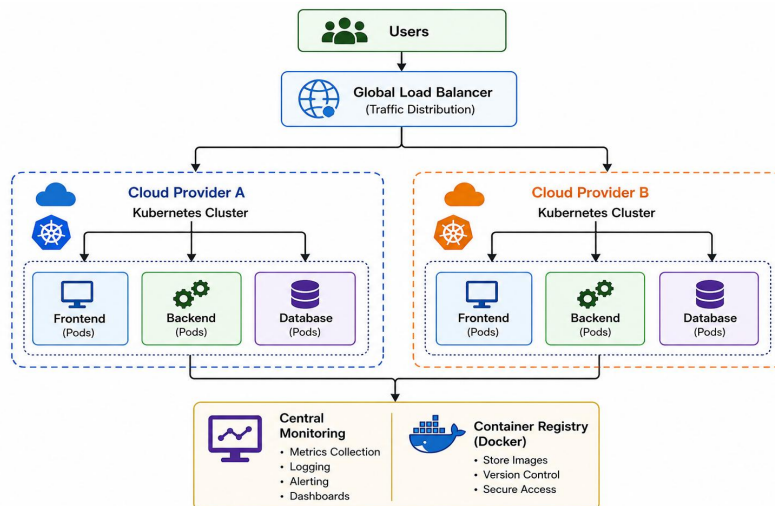
The startup faces the following challenges:

- Inconsistent deployment across different cloud providers.
- Complex container management.
- Difficulty in discovering services across multiple clusters.
- Increased operational complexity.
- Different networking and security configurations.
- Higher monitoring and maintenance effort.

3. Proposed Solution

The organization should adopt **Kubernetes** as the container orchestration platform because it provides automated deployment, scaling, self-healing, and service management across multiple cloud providers.

Proposed Architecture



4. Container Orchestration and Multi-Cloud Strategy

Kubernetes manages container deployment across all cloud providers using a common orchestration platform.

Its features include:

- Automated deployment.
- Rolling updates.
- Self-healing containers.
- Resource scheduling.
- Horizontal pod scaling.
- Centralized management.

A **Container Registry** stores Docker images so that every Kubernetes cluster deploys the same application version, ensuring deployment consistency

5. Service Discovery and Scaling

Service Discovery

Kubernetes automatically assigns DNS names to services.

Instead of using IP addresses, applications communicate using service names such as:

- user-service
- payment-service

- notification-service

This allows containers to communicate even when they are recreated or moved.

Scaling

Kubernetes supports **Horizontal Pod Autoscaling (HPA)**.

When CPU or memory usage increases:

- New container instances are created automatically.

When demand decreases:

- Extra containers are removed.

This ensures efficient resource utilization and high application availability.

6. Benefits of Multi-Cloud Deployment

- High availability.
- Disaster recovery.
- Avoids vendor lock-in.
- Better geographic distribution.
- Improved scalability.
- Increased reliability.
- Business continuity during cloud outages.

7. Risks of Multi-Cloud Deployment

- Increased management complexity.
- Higher networking costs.
- Security policy differences.
- Identity management challenges.
- Data synchronization issues.
- More complex monitoring and troubleshooting.

8. Recommended Solution

The fintech startup should deploy applications using **Docker containers managed by Kubernetes** across multiple cloud providers. A centralized container registry, global load balancer, service discovery, and Horizontal Pod Autoscaling will ensure consistent deployments, efficient scaling, and reliable service communication.

9. Justification

Using Kubernetes with a multi-cloud strategy provides a unified platform for deploying and managing containerized applications. Automated orchestration, service discovery, and auto-scaling reduce

operational complexity while improving scalability, reliability, and availability. Although multi-cloud introduces additional management challenges, proper orchestration and centralized monitoring make it a secure and efficient solution for fintech applications.

Question 3

1. Scenario Understanding

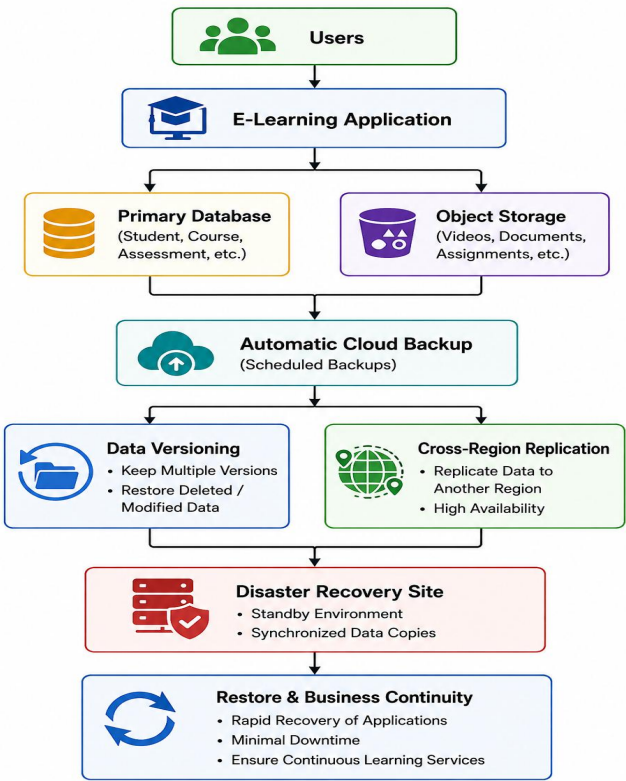
The e-learning platform stores important data such as student records, course materials, assignments, and examination results in the cloud. Due to accidental deletion, critical information has been lost. To prevent permanent data loss and minimize service disruption, the organization requires a reliable backup and disaster recovery (DR) strategy using cloud-native technologies.

2. Challenges / Risks

The organization faces the following risks:

- Permanent loss of academic records.
- Service downtime affecting students and faculty.
- Inconsistent or corrupted data.
- Financial and reputational damage.
- Delays in restoring cloud services.

3. Proposed Cloud Backup & Disaster Recovery Architecture



4. Backup, Replication, and Versioning

Cloud-Native Backup

- Perform automatic daily and incremental backups.
- Store backups securely in cloud storage.
- Schedule periodic backup verification.

Data Replication

- Replicate data to another cloud region or availability zone.
- Maintain synchronized copies to ensure availability during failures.

Versioning

- Enable object versioning so deleted or modified files can be restored.
- Protect against accidental deletion and ransomware attacks.

5. Recovery Time Objective (RTO) and Recovery Point Objective (RPO)

Recovery Time Objective (RTO)

RTO is the **maximum acceptable time required to restore services after a failure.**

To achieve a low RTO:

- Maintain a standby disaster recovery site.
- Automate failover procedures.
- Use replicated databases and cloud infrastructure.

Recovery Point Objective (RPO)

RPO is the **maximum acceptable amount of data loss measured in time.**

To achieve a low RPO:

- Perform frequent incremental backups.
- Enable continuous data replication.
- Use database transaction logs and snapshots.

6. Steps to Ensure Business Continuity

The organization should:

1. Schedule automatic backups.
2. Enable object versioning.
3. Replicate data to another cloud region.
4. Monitor backup health continuously.

5. Perform regular disaster recovery drills.
6. Use automated failover mechanisms.
7. Test backup restoration periodically.
8. Maintain documented recovery procedures.

7. Recommended Solution

The e-learning platform should implement **cloud-native backups**, **cross-region replication**, and **object versioning** to safeguard critical academic data. A disaster recovery site with automated failover and regular backup testing will ensure minimal downtime and rapid restoration of services.

8. Justification

The proposed solution protects against accidental deletion by maintaining multiple recoverable copies of data. Versioning allows restoration of deleted files, while replication ensures service availability even if an entire cloud region fails. By combining cloud-native backup, disaster recovery, and automated failover, the organization can achieve low RTO and RPO values, ensuring business continuity and reliable access to educational services.

Question 4

1. Scenario Understanding

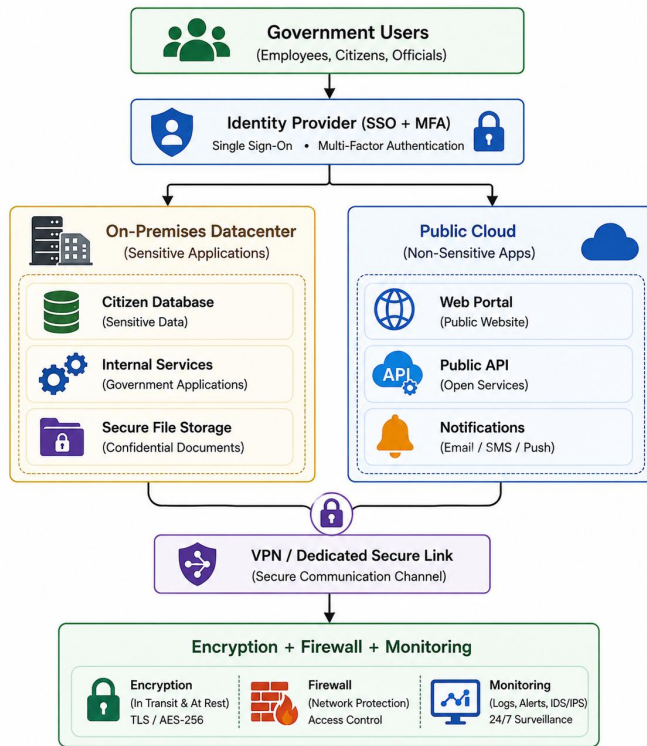
The government agency needs to store **sensitive data** (citizen records, confidential documents, national security information) on-premises to comply with legal and regulatory requirements, while **non-sensitive applications** (public portals, websites, notifications) can be deployed in the public cloud. A **hybrid cloud architecture** allows both environments to work together securely while maintaining compliance and improving scalability.

2. Challenges / Risks

The agency faces the following challenges:

- Protecting sensitive government data.
- Secure communication between on-premises and cloud environments.
- Maintaining compliance with government regulations.
- Managing user identities across both environments.
- Preventing unauthorized access and data breaches.
- Ensuring high availability and disaster recovery.

3. Proposed Hybrid Cloud Architecture



4. Secure Communication

To securely connect the on-premises datacenter with the public cloud:

- Establish a **VPN** or dedicated private connection (e.g., AWS Direct Connect or Azure ExpressRoute).
- Encrypt all communication using **TLS/SSL**.
- Protect the network with firewalls and intrusion detection systems.
- Monitor traffic continuously for suspicious activities.

This ensures that data transferred between environments remains confidential and secure.

5. Identity Management and Encryption

Identity Management

The agency should implement:

- **Single Sign-On (SSO)** for seamless access.
- **Multi-Factor Authentication (MFA)** for stronger security.
- **Role-Based Access Control (RBAC)** to grant permissions based on job roles.
- A centralized **Identity Provider (IdP)** to manage user authentication across both environments.

Encryption

Sensitive data should be:

- Encrypted **at rest** using AES-256.

- Encrypted **in transit** using TLS/SSL.
- Protected with secure key management services.

These measures ensure confidentiality and compliance with government security standards.

6. Challenges in Hybrid Cloud Management

Some common challenges include:

- Complex integration between on-premises and cloud systems.
- Managing identities across multiple environments.
- Ensuring consistent security policies.
- Higher operational and maintenance costs.
- Network latency between environments.
- Compliance with government regulations.

7. Recommended Solution

The government agency should adopt a **hybrid cloud architecture** where sensitive applications remain on-premises while non-sensitive services are hosted in the public cloud. Secure communication should be established through VPN or dedicated private links, while SSO, MFA, RBAC, and encryption ensure secure identity management and data protection.

8. Justification

A hybrid cloud provides the flexibility to use cloud resources without compromising the security of sensitive government data. By combining secure networking, centralized identity management, strong encryption, and continuous monitoring, the agency can achieve regulatory compliance, protect confidential information, and improve scalability and operational efficiency.

Question 5

1. Scenario Understanding

A Software-as-a-Service (SaaS) provider hosts applications that are accessed by users worldwide. To maintain continuous service, the application must remain available even if an entire cloud region experiences an outage. Therefore, a **multi-region deployment** is required to provide high availability, fault tolerance, and business continuity.

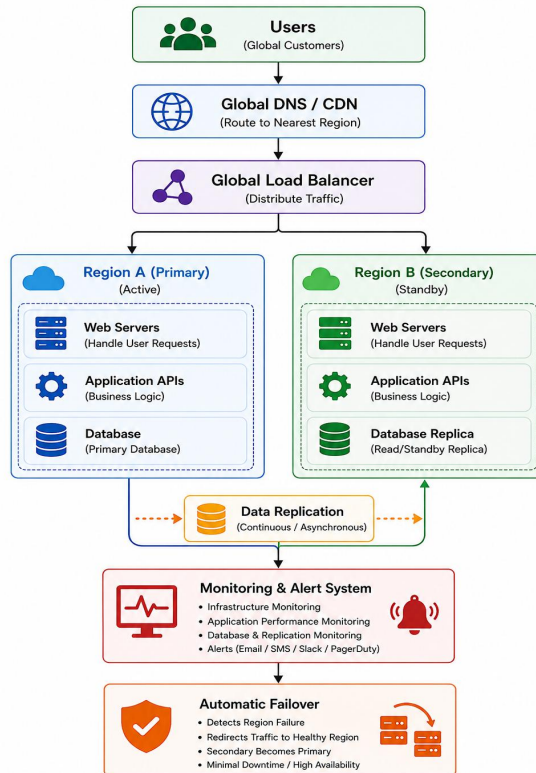
2. Challenges / Risks

The SaaS provider faces the following challenges:

- Complete regional outages due to hardware or network failures.
- Application downtime affecting users.

- Data loss during failures.
- Increased latency for geographically distributed users.
- Delayed detection of system failures.

3. Proposed Multi-Region Cloud Architecture



4. Data Replication and Failover

Data Replication

The database should continuously replicate data from the primary region to the secondary region.

Benefits:

- Maintains identical copies of application data.
- Prevents data loss.
- Ensures quick recovery after failures.
- Supports disaster recovery.

Failover Mechanism

Automatic failover continuously monitors the health of the primary region.

If Region A fails:

- Traffic is redirected automatically to Region B.

- The replicated database becomes the active database.
- Users continue accessing the application with minimal interruption.

This minimizes downtime and improves service availability.

5. Monitoring and Alerting

The SaaS provider should implement continuous monitoring using cloud monitoring tools.

Monitoring should include:

- CPU utilization.
- Memory usage.
- Network traffic.
- Application response time.
- Database health.
- Service availability.

Alerts should notify administrators through:

- Email.
- SMS.
- Mobile notifications.
- Incident management tools.

Automatic health checks should trigger failover if a region becomes unavailable.

6. Recommended Solution

The SaaS provider should deploy the application across **multiple cloud regions** with:

- Global Load Balancer.
- Continuous database replication.
- Automatic failover.
- Cloud monitoring and alerting.
- Auto Scaling in each region.

This ensures uninterrupted service even if one cloud region experiences an outage.

7. Justification

A multi-region architecture significantly improves **high availability**, **fault tolerance**, and **business continuity** by eliminating single points of failure. Continuous data replication protects against data loss, while automatic failover ensures users experience minimal downtime. Monitoring and proactive alerting enable rapid detection and response to failures, making the solution highly reliable for SaaS applications.

